

Sensitivity analysis

Kent Andersen

We consider a linear program (P) in standard form:

$$\begin{aligned} & \text{Minimize } Z_P = c^T x \\ & \text{Subject to} \\ & \quad Ax = b \quad (P) \\ & \quad x \geq 0_n \end{aligned}$$

and its dual (D):

$$\begin{aligned} & \text{Maximize } Z_D = b^T y \\ & \text{Subject to} \\ & \quad A^T y \leq c \quad (D) \end{aligned}$$

We assume we have solved the problems (P) and (D), and that $B \subseteq N := \{1, \dots, n\}$ is an optimal basis. We will use B to perform sensitivity analysis on the parameters defining (P). In other words, we will use B to solve modified versions (P') and (D') of the problems (P) and (D) for modified data A' , b' and c' (whenever B is useful for this purpose). This is best demonstrated by examples. We consider the following linear program (P):

$$\begin{aligned} & \text{Minimize } Z_P = -3x_1 - 5x_2 \\ & \text{Subject to} \\ & \quad x_1 + x_3 = 4 \quad (y_1) \\ & \quad 2x_2 + x_4 = 12 \quad (y_2) \quad (P) \\ & \quad 3x_1 + 2x_2 + x_5 = 18 \quad (y_3) \\ & \quad x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

and the corresponding dual linear program (D):

$$\text{Maximize } Z_D = 4y_1 + 12y_2 + 18y_3$$

Subject to

$$\begin{array}{rcl} y_1 + 3y_3 & \leq & -3 \quad (x_1) \\ 2y_2 + 2y_3 & \leq & -5 \quad (x_2) \\ y_1 & \leq & 0 \quad (x_3) \\ y_2 & \leq & 0 \quad (x_4) \\ p_3 & \leq & 0 \quad (x_5) \end{array} \quad (D)$$

The basis $B = \{1, 2, 3\} \subseteq N = \{1, \dots, 5\}$ is optimal for the problems (P) and (D). The optimal basic solution for (P) is $\bar{x} = (2, 6, 2, 0, 0)$. The optimal basis matrix A_B , and its inverse $(A_B)^{-1}$, are given by:

$$A_B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 2 & 0 \end{pmatrix} \quad \text{and} \quad (A_B)^{-1} = \begin{pmatrix} 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & \frac{1}{2} & 0 \\ 1 & \frac{1}{3} & -\frac{1}{3} \end{pmatrix}.$$

The optimal dual basic solution $\bar{p}$ associated with B is the unique solution to the system $(A_j)^T y = c_j$ for $j \in B$, where A_j denotes the j^{th} column of A , or equivalently the unique solution to the system $y_1 + 3y_3 = -3$, $2y_2 + 2y_3 = -5$ and $y_1 = 0$. This gives $\bar{y}^T = (0, -\frac{3}{2}, -1)$.

The reduced costs on the variables are the values of the slack variables $\bar{s}$ in the dual constraints. The slack variables in the dual constraints corresponding to x_1 , x_2 and x_3 are zero, since these variables are basic variables. Hence we have $\bar{s}_1 = \bar{s}_2 = \bar{s}_3 = 0$. For the dual constraints corresponding to the non-basic variables x_4 and x_5 the values of the dual slack variables are given by $\bar{s}_4 = c_4 - (A_4)^T \bar{y} = 0 - (-\bar{y}_2) = \frac{3}{2}$ and $c_5 - (A_5)^T \bar{y} = 0 - (-1) = 1$. Since all reduced costs (dual slack variables) are non-negative, the basis B is an optimal basis.

The updated columns for the non-basic variables x_4 and x_5 are given by $\bar{A}_4 = (A_B)^{-1} A_4$ and $\bar{A}_5 = (A_B)^{-1} A_5$:

$$\bar{A}_4 = \begin{pmatrix} -\frac{1}{3} \\ \frac{1}{2} \\ \frac{1}{3} \end{pmatrix} \quad \text{and} \quad \bar{A}_5 = \begin{pmatrix} \frac{1}{3} \\ 0 \\ -\frac{1}{3} \end{pmatrix}.$$

The information available is summarized in the simplex tableau below. We now consider various changes in the data A , b and c .

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	Right hand side
Z_P	-1	0	0	0	$\frac{3}{2}$	1	36
x_1	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
x_2	0	0	1	0	$\frac{1}{2}$	0	6
x_3	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2

Example 1 (*Change of right hand side*)

Suppose the second equality $2x_2 + x_4 = 12$ is obtained from $2x_2 \leq 12$ by adding a slack variable x_4 to the inequality. Suppose the number "12" refers to a capacity of a resource in a production company. What happens if the capacity of this resource is increased by, say, δ ? In other words, what happens if the inequality $2x_2 \leq 12$ is changed to $2x_2 \leq 12 + \delta$? Let (P') and (D') denote the new linear programs, and let b' denote the new right-hand-side vector ($A' = A$ and $c' = c$).

First observe that the basis matrix A_B remains non-singular (since A_B is not changed). Therefore B remains a basis for (P') . Secondly, the dual basic solution corresponding to B is given as the unique solution $\bar{y}$ to the system $(A_j)^T y = c_j$ for $j \in B$. Hence, since $A' = A$ and $c' = c$, the "old" dual basic feasible solution $\bar{y}$ remains a dual basic feasible solution for (D') .

It follows that, if B is a feasible basis for (P') , then B is also an optimal basis for (P') and (D') . To compute the new basic solution $\bar{x}'$, we need to solve the system $\sum_{j \in B} A_j x_j = b'$, or in other words:

$$\begin{aligned}
A_B \begin{pmatrix} \bar{x}'_1 \\ \bar{x}'_2 \\ \bar{x}'_3 \end{pmatrix} &= b + \begin{pmatrix} 0 \\ \delta \\ 0 \end{pmatrix} && \Rightarrow \\
\begin{pmatrix} \bar{x}'_1 \\ \bar{x}'_2 \\ \bar{x}'_3 \end{pmatrix} &= (A_B)^{-1} b + (A_B)^{-1} \begin{pmatrix} 0 \\ \delta \\ 0 \end{pmatrix} && \Rightarrow \\
\begin{pmatrix} \bar{x}'_1 \\ \bar{x}'_2 \\ \bar{x}'_3 \end{pmatrix} &= \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix} + \delta \begin{pmatrix} -\frac{1}{3} \\ \frac{1}{2} \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 2 - \frac{1}{3}\delta \\ 6 + \frac{1}{2}\delta \\ 2 + \frac{1}{3}\delta \end{pmatrix}.
\end{aligned}$$

Observe that, as long as $-6 \leq \delta \leq 6$, the new basic solution $\bar{x}'$ is feasible for (P') . It follows that B is an optimal basis for (P') if $-6 \leq \delta \leq 6$.

What is the objective value of the basic solution in the case when $-6 \leq \delta \leq 6$? This change is given by $(b')^T \bar{y} = b^T \bar{y} + [0, \delta, 0] \bar{y} = b^T \bar{y} + \delta \bar{y}_2$. We have $\bar{y}_2 = -\frac{3}{2}$, which shows that the cost can be decreased by $\frac{3}{2}$ pr. unit of the second resource. This is why the optimal dual variables are also called shadow prices (implicit prices on the resources).

Note that when $\delta \notin [-6, 6]$, a basic variable x_j for $j \in B$ becomes negative. Hence, in this case, a feasible basis for (D') is available, which is not feasible for (P') . It is therefore natural to re-optimize (P) and (D) with the dual simplex method.

Example 2 (Changing coefficients on non-basic variables)

Now consider a non-basic variable $\bar{j} \in N \setminus B$, and suppose we change $c_{\bar{j}}$ to $c'_{\bar{j}}$ and $A_{\bar{j}}$ to $A'_{\bar{j}}$. Observe that B is still a feasible basis for (P') after the change (since $b' = b$ and $A'_j = A_j$ for all $j \in B$). Also, since $c'_j = c_j$ for every $j \in B$, the dual basic solution corresponding to B remains the same. Finally, since $A'_j = A_j$ and $c'_j = c_j$ for every non-basic variable x_j different from $x_{\bar{j}}$, we have that $\bar{y}$ satisfies all constraints of (D') except perhaps the inequality $(A'_{\bar{j}})^T \bar{y} \leq c'_{\bar{j}}$.

Hence the question is whether $\bar{y}$ satisfies the inequality $(A'_{\bar{j}})^T \bar{y} \leq c'_{\bar{j}}$. If yes, then $\bar{x}$ and $\bar{y}$ remain optimal, and if no, the reduced cost (dual slack) for the variable $x_{\bar{j}}$ is negative and given by $c'_{\bar{j}} - (A'_{\bar{j}})^T \bar{y} < 0$. The variable $x_{\bar{j}}$ therefore becomes a candidate (in fact the only candidate) for entering the basis. Hence, in the case when $c'_{\bar{j}} - (A'_{\bar{j}})^T \bar{y} < 0$, the primal simplex algorithm can be used to re-optimize (P') and (D') .

In the example, suppose the cost coefficient c_4 on x_4 is reduced from $c_4 = 0$ to $c'_4 = -1$ (making x_4 more attractive). Does $\bar{y} = (0, -1, -\frac{3}{2})$ satisfy $(A_4)^T \bar{p} \leq c'_4$? The answer is yes, since $(A'_4)^T \bar{y} = (A_4)^T \bar{p} = -\frac{3}{2} \leq -1 = c'_4$. Hence, the optimal solutions $\bar{x} = (2, 6, 2, 0, 0)$ and $\bar{y} = (0, -1, -\frac{3}{2})$ for (P) and (D) are also optimal for (P') and (D') . In other words, the reduction of the cost coefficient of x_4 is not sufficient to make x_4 sufficiently attractive to enter the basis.

Example 3 (Adding a new variable)

Suppose we add a new variable x_{new} to the problem. The variable x_{new} could represent a new product that could potentially be produced.

Minimize $Z_P = -3x_1 - 5x_2 - 4x_{new}$

Subject to

$$\begin{aligned} x_1 + x_3 + 2x_{new} &= 4 & (y_1) \\ 2x_2 + x_4 + 3x_{new} &= 12 & (y_2) \quad (P') \\ 3x_1 + 2x_2 + x_5 + x_{new} &= 18 & (y_3) \\ x_1, x_2, x_3, x_4, x_5, x_{new} &\geq 0 \end{aligned}$$

We have $c_{new} = -4$ and $A_{new} = (2, 3, 1)^T$. The basis B is a feasible basis for the modified problem (P') . The question is whether B is still optimal. This question is answered by checking whether $\bar{y}$ satisfies the new dual constraint $(A_{new})^T \bar{y} \leq c_{new}$. Since $(A_{new})^T \bar{y} = -\frac{11}{2}$ and $c_{new} = -4$, the new dual constraint is indeed satisfied. This implies that the reduced cost (or dual slack) for the new variable x_{new} is positive, and therefore B is also optimal for (P') . If the reduced cost on x_{new} was negative, the primal simplex method could be applied from B by letting x_{new} enter the basis. Therefore, if the variable x_{new} represents a new product, the conclusion is that it is not profitable to produce this product.

Example 4 (Changing the coefficients on a basic variable)

Let $k \in B$ be a basic variable. Suppose we change c_k to c'_k and A_k to A'_k . Since this changes a column of A_B , the matrix A_B may no longer be non-singular, and therefore B may not be a basis for the problem (P') . However, it is still possible to use the current information from B . We use $B = \{i_1, \dots, i_m\}$ to index the variables in B , and the basic solution corresponding to B is denoted $\bar{x}$.

Every solution to (P) satisfies the equations $Ax = b$. For the modified problem (P') , every solution satisfies $A'_k x_k + \sum_{j \in N \setminus \{k\}} A_j x_j = b$. Multiplying these new equations with the OLD basis-inverse $(A_B)^{-1}$ gives:

$$\begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = (A_B)^{-1} b = \bar{A}'_k x_k + \sum_{j \in N \setminus \{k\}} \bar{A}_j x_j, \quad (1)$$

where $\bar{A}'_k := (A_B)^{-1} A'_k$ and $\bar{A}_j := (A_B)^{-1} A_j$ for $j \in N \setminus \{k\}$. This is almost the old equations $\sum_{j \in N} \bar{A}_j x_j = (A_B)^{-1} b$ except for changing $\bar{A}_k$ to $\bar{A}'_k$.

If we multiply equations (1) with $-(c^B)^T = -(c_{i_1}, c_{i_2}, \dots, c_{i_m})$ on both sides we obtain the equation:

$$-\sum_{s=1}^m c_{i_s} \bar{x}_{i_s} = -c^T \bar{x} = -x_k \cdot (c^B)^T \bar{A}'_k - \sum_{j \in N \setminus \{k\}} x_j \cdot (c^B)^T \bar{A}_j,$$

which must be satisfied by every solution x to (P') . If we add this equation to the new cost equation $0 = -Z_{P'} + c'_k x_k + \sum_{j \in N \setminus \{k\}} c_j x_j$ we get:

$$-c^T \bar{x} = -Z_{P'} + \bar{s}'_k x_k + \sum_{j \in N \setminus \{k\}} \bar{s}_j x_j \quad (2)$$

where $\bar{s}'_k := c'_k - (c^B)^T \bar{A}'_k$ and $\bar{s}_j := c_j - (c^B)^T \bar{A}_j$ for $j \in N \setminus \{k\}$. Observe that the numbers $\bar{s}_j$ for $j \in N \setminus \{k\}$ are the reduced costs (dual slacks) for the old problem (P) . Hence, to compute the new cost equation (2) from the old cost equation $-c^T \bar{x} = -Z_P + \sum_{j \in N} \bar{s}_j x_j$ all that is needed is to update $\bar{s}_k$ to $\bar{s}'_k$.

The system (1)-(2) almost gives the simplex tableau for (P') , except that the column corresponding to x_k is no longer a unit vector. However, if B is still a basis for the modified problem (P') , then it is possible to make this column a unit vector with row operations. One can then create a simplex tableau for (P') for the basis B .

We now illustrate this approach on the example. Recall that the optimal basis is $B = \{1, 2, 3\}$, and that the optimal simplex tableau is given by:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	Right hand side
Z_P	-1	0	0	0	$\frac{3}{2}$	1	36
x_1	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
x_2	0	0	1	0	$\frac{1}{2}$	0	6
x_3	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2

Suppose we change the coefficients on the basic variable x_2 :

$$c_2 = -5 \rightarrow c'_2 = -3 \quad \text{and} \quad A_2 = (0, 2, 2)^T \rightarrow A'_2 = (0, 2, 3)^T.$$

$$\text{Hence } k = 2, \bar{A}'_k = (A_B)^{-1} A'_k = (\frac{1}{3}, 1, -\frac{1}{3})^T \text{ and } \bar{s}'_2 = c'_2 - (c^B)^T \bar{A}'_k = 3.$$

With this information we can now write (1)-(2) as:

$$36 = -Z_{P'} + 3x_2 + \frac{3}{2}x_4 + x_5$$

$$\begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} x_1 + \begin{pmatrix} \frac{1}{3} \\ 1 \\ -\frac{1}{3} \end{pmatrix} x_2 + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} x_3 + \begin{pmatrix} -\frac{1}{3} \\ \frac{1}{2} \\ \frac{1}{3} \end{pmatrix} x_4 + \begin{pmatrix} \frac{1}{3} \\ 0 \\ -\frac{1}{3} \end{pmatrix} x_5$$

The above equations can be written in tableau form as follows:

Basic Variable	$Z_{P'}$	x_1	x_2	x_3	x_4	x_5	Right hand side
$Z_{P'}$	-1	0	3	0	$\frac{3}{2}$	1	36
x_1	0	1	$\frac{1}{3}$	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
x_2	0	0	1	0	$\frac{1}{2}$	0	6
x_3	0	0	$-\frac{1}{3}$	1	$\frac{1}{3}$	$-\frac{1}{3}$	2

Observe that this is exactly the "old" tableau except that the column corresponding to x_2 has changed. Therefore, in order to compute the above table, we only needed to update the column corresponding to x_2 (the vector $\bar{A}'_2$ and the number $\bar{s}'_2$).

In order to make this a tableau for the modified problem (P'), we simply need to perform row operations on the above table to make the column corresponding to x_2 a unit vector. For this first multiply the row corresponding to x_2 with (-3) , and add the result to the row corresponding to $Z_{P'}$. This gives the following table.

Basic Variable	$Z_{P'}$	x_1	x_2	x_3	x_4	x_5	Right hand side
$Z_{P'}$	-1	0	0	0	0	1	18
x_1	0	1	$\frac{1}{3}$	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
x_2	0	0	1	0	$\frac{1}{2}$	0	6
x_3	0	0	$-\frac{1}{3}$	1	$\frac{1}{3}$	$-\frac{1}{3}$	2

Next we multiply the row corresponding to x_2 with $\frac{1}{3}$, and add the result to the row corresponding to x_3 . This gives the following table.

Basic Variable	$Z_{P'}$	x_1	x_2	x_3	x_4	x_5	Right hand side
$Z_{P'}$	-1	0	0	0	0	1	18
x_1	0	1	$\frac{1}{3}$	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
x_2	0	0	1	0	$\frac{1}{2}$	0	6
x_3	0	0	0	1	$\frac{1}{2}$	$-\frac{1}{3}$	4

Finally we multiply the row corresponding to x_2 with $-\frac{1}{3}$, and add the result to the row corresponding to x_1 . We then get the final table, which is the simplex tableau for the modified problem (P') corresponding to the basis $B = \{1, 2, 3\}$.

Basic Variable	$Z_{P'}$	x_1	x_2	x_3	x_4	x_5	Right hand side
$Z_{P'}$	-1	0	0	0	0	1	18
x_1	0	1	0	0	$-\frac{1}{2}$	$\frac{1}{3}$	0
x_2	0	0	1	0	$\frac{1}{2}$	0	6
x_3	0	0	0	1	$\frac{1}{2}$	$-\frac{1}{3}$	4

The new basic solution for the modified problem (P') from the basis $B = \{1, 2, 3\}$ is then $\bar{x} = (0, 6, 4, 0, 0)^T$. Furthermore, the reduced costs on the non-basic variables x_4 and x_5 are given by $\bar{s}_4 = 0$ and $\bar{s}_5 = 1$. Since the reduced cost (dual slacks) are non-negative, this solution is optimal for the modified problem (P').

In this example, the old basis B remained a basis for the modified problem (P'). In fact, the old basis remained an optimal basis. In general, the old basis may be neither primal feasible nor dual feasible for (P'). However, if the old basis stays either primal feasible or dual feasible, there is a natural starting basis for either the primal simplex method or the dual simplex method.